

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2023 - 2024)

**JEE(Advanced)****PRE-SCORE****15-10-2023****JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)**

Time : 3 Hours

**PAPER-2 (OPTIONAL)**

Maximum Marks : 180

**IMPORTANT NOTE :** Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

**READ THE INSTRUCTIONS CAREFULLY****GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

**OPTICAL RESPONSE SHEET :**

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

**DARKENING THE BUBBLES ON THE ORS :**

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take  **$g = 10 \text{ m/s}^2$**  unless otherwise stated.

**QUESTION PAPER FORMAT :**

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

**SOME USEFUL CONSTANTS**

**Atomic No.** : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

**Atomic masses** : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- |                                    |                                                              |
|------------------------------------|--------------------------------------------------------------|
| • Boltzmann constant               | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$                   |
| • Coulomb's law constant           | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$                   |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$  |
| • Speed of light in vacuum         | $c = 3 \times 10^8 \text{ ms}^{-1}$                          |
| • Stefan-Boltzmann constant        | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$                        |
| • Permeability of vacuum           | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$                |
| • Permittivity of vacuum           | $\epsilon_0 = \frac{1}{\mu_0 c^2}$                           |
| • Planck constant                  | $h = 6.63 \times 10^{-34} \text{ J-s}$                       |

Space for Rough Work

**PART-1 : PHYSICS**

**SECTION-I (i) : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

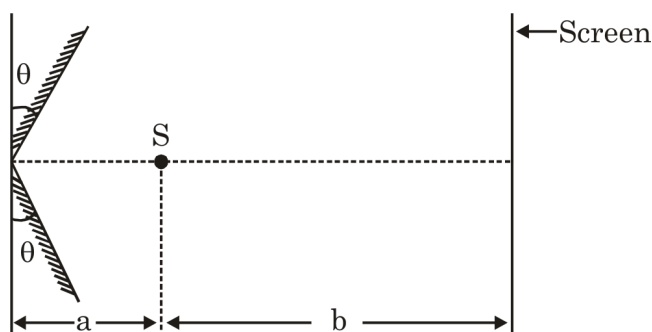
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. An air filled capacitor plate has a plate separation of  $0.885 \pm 0.009$  mm. The area of the plates is  $0.040 \pm 0.001$  m<sup>2</sup>. It is connected to a voltage source of voltage  $10.0 \pm 0.1$  V. Choose the correct statement(s)
  - (A) The charge has a percentage error of approximately 4.5%.
  - (B) The energy stored has a percentage error of approximately 5.5%
  - (C) The force between the plates has a percentage error of approximately 5.5%.
  - (D) If we insert a dielectric of dielectric constant  $k = 4$  which fills the region between them completely, the random error in energy increases.

2. Ram performs optical bench experiment to find out the focal length of a concave lens. For this purpose, he first takes a convex lens and measures the position of object needle and image needle. Then he combines convex lens and concave lens and again measures the object needle and image needle position. The optical bench has no bench error. The readings obtained by Ram in the two experiments are shown in the table below.

|                         | Object needle | Lens    | Image needle |
|-------------------------|---------------|---------|--------------|
| Convex lens             | 37.1 cm       | 50.6 cm | 77.6 cm      |
| Convex and Concave lens | 37.1 cm       | 50.6 cm | 91.1 cm      |

- (A) The focal length of the convex lens is 9.0 cm.  
 (B) The focal length of the concave lens is 81 cm in modulus.  
 (C) The error in focal length of convex lens is approximately 0.1 cm.  
 (D) The error in focal length of concave lens is approximately 1.0 cm.
3. Consider an ideal solenoid having radius of cross section  $R$ . There are  $n$  turns/length. It carries a current of  $I_0 \sin(\omega t)$ . Consider a cylinder which is coaxial with the solenoid and has a length  $\ell$  and radius  $r$  ( $r < R$ ). (Neglect magnetic field due to induced electric field)
- (A) The rate of energy flow through its flat face is zero.  
 (B) At  $t = \frac{\pi}{4\omega}$ , energy is flowing outwards from the curved surface of the cylinder.  
 (C) The rate of energy flow is proportional to  $\mu_0 n^2 i_0^2 r^2 \omega \ell$ .  
 (D) At  $t = \frac{\pi}{4\omega}$ , rate of energy flow is maximum.
4. Two plane mirrors, a source  $S$  of light, emitting wavelength of  $\lambda_1 = 4000\text{\AA}$  and  $\lambda_2 = 5600\text{\AA}$  and a screen are arranged as shown in figure. The angle  $\theta$  shown is 0.05 radian and distance  $a$  and  $b$  are 1 cm and 38 cm respectively.



Choose the correct options.

- (A) fringe width of  $\lambda = 4000\text{\AA}$  on screen is 0.08mm  
 (B) Angular fringe width of  $\lambda = 5600\text{\AA}$  on screen is  $2.8 \times 10^{-4}$  rad  
 (C) Distance of first black line from central maxima is 0.25 mm  
 (D) The distance between two dark lines nearest to central maxima is 0.5 mm.

5. A plane electromagnetic harmonic wave of frequency  $600 \times 10^{12}$  Hz (green light), propagating in the positive x direction in vacuum, has an electric field amplitude of 42.43 V/m. The wave is linearly polarized such that the plane of vibration of the electric field is at  $45^\circ$  to the xz plane.

(A) The equation for  $\vec{E}$  is  $\vec{E} = (30\text{V/m}) \sin\left(2\pi (600 \times 10^{12}) \left(t - \frac{x}{3 \times 10^8}\right)\right) (\hat{j} + \hat{k})$

(B) The equation for  $\vec{B}$  is  $\vec{B} = (10^{-7}\text{T}) \sin\left(2\pi (600 \times 10^{12}) \left(t - \frac{x}{3 \times 10^8}\right)\right) (\hat{j} + \hat{k})$

(C) For equation for  $\vec{B}$  is  $\vec{B} = (10^{-7}\text{T}) \sin\left(2\pi (600 \times 10^{12}) \left(t - \frac{x}{3 \times 10^8}\right)\right) (-\hat{j} + \hat{k})$

(D) For equation for  $\vec{S}$  is  $\vec{S} = \left(\frac{15}{\pi}\text{W/m}^2\right) \sin^2\left(2\pi (600 \times 10^{12}) \left(t - \frac{x}{3 \times 10^8}\right)\right) \hat{i}$

6. A thin film of magnesium fluoride ( $\mu = 1.38$ ) is used to coat a camera lens ( $\mu = 1.56$ ). The thickness of coating which will not allow any strong reflection for the wavelength 552 nm is/are :

(A) 1000 nm

(B) 500 nm

(C) 300 nm

(D) 100 nm

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +3 If ONLY the correct option is chosen.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered)

*Negative Marks* : -1 In all other cases

7. During space exploration, ISRO found a new planet similar to Earth. Experiments are conducted by four astronauts using simple pendulum to measure the value of  $g$  on this new planet. The observations are shown in table below.

| Astronaut | Length of pendulum (cm) | number of oscillations (n) | Total Time (sec) |
|-----------|-------------------------|----------------------------|------------------|
| (1)       | 10                      | 12                         | 40               |
| (2)       | 20                      | 5                          | 60               |
| (3)       | 5                       | 10                         | 30               |
| (4)       | 10                      | 8                          | 20               |

Least count of length is 0.1 cm and least count of time is also 0.1 sec. List-I given about four astronauts and List-II gives fractional error in the measurement of  $g$ .

| List-I |             | List-II |                 |
|--------|-------------|---------|-----------------|
| (P)    | Astronaut-1 | (1)     | $\frac{3}{200}$ |
| (Q)    | Astronaut-2 | (2)     | $\frac{1}{120}$ |
| (R)    | Astronaut-3 | (3)     | $\frac{2}{75}$  |
| (S)    | Astronaut-4 | (4)     | $\frac{1}{50}$  |

- (A)  $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$
- (B)  $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$
- (C)  $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$
- (D)  $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

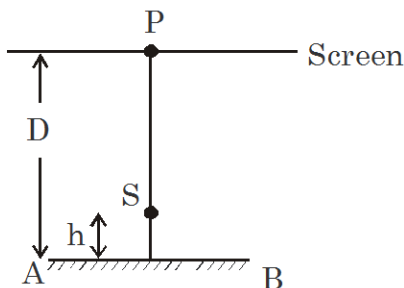
8. Two plane wavefronts  $S_1 = (2\text{mm}) \sin (340\pi t - \pi x)$  and  $S_2 = 2\text{mm} \sin (\pi y - 340\pi t)$  are travelling along x and y direction respectively. They represent 2 sound waves in same medium. At different point in space, comment on motion of particles of the medium and intensity.

| List-I |         | List-II |                                        |
|--------|---------|---------|----------------------------------------|
| (P)    | (0,0)   | (1)     | Particles will execute SHM             |
| (Q)    | (2,1)   | (2)     | Particles will undergo circular motion |
| (R)    | (0.5,2) | (3)     | Intensity will be maximum              |
| (S)    | (3,5)   | (4)     | Intensity will be minimum              |

- (A)  $P \rightarrow 1,3; Q \rightarrow 1,4; R \rightarrow 2; S \rightarrow 1,3$   
 (B)  $P \rightarrow 1,3; Q \rightarrow 1,4; R \rightarrow 3; S \rightarrow 2,4$   
 (C)  $P \rightarrow 2,4; Q \rightarrow 1; R \rightarrow 1,4; S \rightarrow 1,2$   
 (D)  $P \rightarrow 3; Q \rightarrow 2,4; R \rightarrow 1,3; S \rightarrow 3,4$

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

A point source S emitting monochromatic light having wavelength  $\lambda$  is placed at a very small height  $h = 2\lambda$  above flat reflecting surface AB as shown in diagram. The intensity of the reflected light is 36% of incident intensity. Interference fringes are observed on a screen placed parallel to the reflecting surface at a large distance  $D (>>h)$  from it. The nature of interference fringes formed on screen is circular.



List I gives some physical quantities and List II gives some possible values of these physical quantities ( $I_0$  is incident intensity)

| List-I |                         | List-II |                         |
|--------|-------------------------|---------|-------------------------|
| (P)    | Intensity of maxima     | (1)     | $\sqrt{143} D$          |
| (Q)    | Intensity of minima     | (2)     | $2.56 I_0$              |
| (R)    | Radius of first maxima  | (3)     | $0.16 I_0$              |
| (S)    | Radius of second maxima | (4)     | $\sqrt{63} D$           |
|        |                         | (5)     | $\frac{\sqrt{55} D}{3}$ |
|        |                         | (6)     | $\sqrt{15} D$           |

(A)  $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 6$

(B)  $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 5$

(C)  $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 4$

(D)  $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 6$

10. A standing wave, formed in an organ pipe aligned along the x-axis, is given by

$$S = (5 \times 10^{-6} \text{ m}) \sin(4\pi x) \cos(1320\pi t)$$

Where x is in m and t is in seconds.

| List-I |                                            | List-II |                             |
|--------|--------------------------------------------|---------|-----------------------------|
| (P)    | Magnitude of change in pressure is maximum | (1)     | $x = \frac{1}{4} \text{ m}$ |
| (Q)    | Magnitude of change in pressure is minimum | (2)     | $x = \frac{1}{2} \text{ m}$ |
| (R)    | Maximum amplitude of vibration             | (3)     | $x = 1 \text{ m}$           |
| (S)    | Minimum amplitude of vibration             | (4)     | $x = \frac{1}{8} \text{ m}$ |
|        |                                            | (5)     | $x = \frac{3}{8} \text{ m}$ |

- (A)  $P \rightarrow 1, 2, 3; Q \rightarrow 4, 5; R \rightarrow 4, 5; S \rightarrow 1, 2, 3$
- (B)  $P \rightarrow 1, 3; Q \rightarrow 3, 5; R \rightarrow 2, 3, 4; S \rightarrow 1, 2, 3, 4, 5$
- (C)  $P \rightarrow 2, 3; Q \rightarrow 2, 4; R \rightarrow 2, 3, 5; S \rightarrow 3, 4, 5$
- (D)  $P \rightarrow 1, 3, 4; Q \rightarrow 1, 3; R \rightarrow 2, 3; S \rightarrow 2, 3, 4, 5$

## SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +3 If ONLY the correct numerical value is entered.

*Zero Marks* : 0 In all other cases.

1. In an experiment to measure the Planck constant,  $h$ , the following results were obtained, in units of  $10^{-34}$  Js : 6.65, 6.67, 6.57, 6.61, 6.72, 6.60, 6.71, 6.66, 6.64, 6.67. The percentage error in measurement of Planck constant is
2. A screw gauge is used to measure the thickness of a thin sheet of copper. The pitch of screw gauge is 0.2 mm and total number of division on circular scale is 200. When two jaws are brought in contact then 160<sup>th</sup> division on circular scale is exactly coinciding with the main scale line, and that the zero of main scale is not visible. When thickness of sheet is measured with screw gauge then main scale reading is 0.6 mm and 100<sup>th</sup> division coincide with the main scale line. Find the thickness of copper sheet (in mm).
3. The plane flies horizontally at an altitude of  $H = 4$  km at supersonic speed. The sound reaches an observer on ground 10 seconds after the plane crosses him from above. The speed of sound is 320 m/s. The speed of plane (in m/s) is
4.  $S_1$  and  $S_2$  are identical sources of sound of a single frequency. They are equidistant from O. When  $S_1$  alone is sounding, the amplitude reaching O is A. What is the increase in acoustic intensity in decibels when  $S_2$  is turned on ?
5. A radar antenna is located atop a high cliff on the edge of a lake. This antenna operates on a wavelength of 400 m. Venus rises above the horizon and is tracked by the antenna. The first minimum in the signal reflected off the surface of Venus is recorded when Venus is  $35^\circ$  above the horizon. The height of cliff (in m) is : ( $\sin 35^\circ = 0.58$ ,  $\sin 20^\circ = 0.34$ )

6. Light of wavelength 400 nm is incident normally on a plano-convex glass lens placed on a horizontal glass surface. The radius of curvature of the curved surface is 5m. The curved face touches glass surface. The radius of third bright fringe (in mm) is
7. Polarized light of intensity  $I_0$  is incident on a pair of Polaroid sheets. Let  $\theta_1$  and  $\theta_2$  be the angle between the incident amplitude and the axes of the first and second sheets, respectively. If  $\theta_1 = 30^\circ$  and  $\theta_2 = 60^\circ$ , the intensity of transmitted light is  $\alpha I_0$ , the value of  $\alpha$  is
8. In single slit fraunhofer diffraction pattern having a slit width  $a = \lambda$ , resultant amplitude at the centre of central maxima is  $A_0$  and it is  $A$  at an angular position of  $30^\circ$ . Then  $\frac{A}{A_0}$  will be

**PART-2 : CHEMISTRY**  
**SECTION-I (i) : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

- 
1. Which of the following is/are correct statement/s about alkali metals.
    - (A) Reaction with water becomes increasingly violent on descending the group
    - (B) The reaction  $M^{\oplus} + e^{\ominus} \rightarrow M$  has largest negative  $E^{\ominus}$  value for lithium and hence the largest positive  $\Delta G^{\ominus}$  value among alkali metals
    - (C)  $KO_2$  is used in space capsules, submarines and breathing masks because it produces dioxygen and remove carbon dioxide
    - (D) All carbonate salt of alkali metal on heating give  $CO_2$
  2. In which of the following reaction/s hydrogen obtain as a product.
    - (A)  $Na[BH_4] + I_2 \rightarrow$
    - (B)  $BF_3 + LiAlH_4 \rightarrow$
    - (C)  $BF_3 + NaH \rightarrow$
    - (D)  $B_2H_6 + H_2O \rightarrow$

3. Which of the following is/are correct statements about allotropes of carbon.
- (A)  $C_{60}$  contain total twenty 5-membered ring and it contain aromatic character
- (B) Graphite is composed of flat two dimensional sheets of carbon atoms each sheet is a hexagonal net of C atoms
- (C) Graphite is thermodynamically more stable than diamond
- (D) Fullerene is soluble in organic solvent
4. Which of the following reaction give a product which contain N-N bond.
- (A)  $Mg_3N_2 \xrightarrow{\Delta}$
- (B)  $NH_4NO_2 \xrightarrow{\Delta}$
- (C)  $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$
- (D)  $(NH_4)_2SO_4 \xrightarrow{\Delta}$
5. In which of the following series all the species contains bond between same atom.
- (A)  $H_2SO_5$ ,  $H_2S_2O_8$ ,  $H_2S_2O_3$
- (B)  $C_3O_2$ ,  $S_3O_9$ ,  $N_2O_5$
- (C)  $N_2O$ ,  $N_2O_4$ ,  $N_2O_3$ (unsymmetrical)
- (D)  $H_2S_2O_4$ ,  $H_2S_4O_6$ ,  $H_2S_2O_6$
6. Choose correct statement about paramagnets.
- (A) Permagnates can be prepared by treating a solution of  $Mn^{2+}$  excess with very strong oxidizing agent such as  $PbO_2$ , or sodium bismuthate
- (B) Colour in  $KMnO_4$  is due to d-d transition
- (C) When  $MnO_4^-$  react with  $I^-$  in acidic medium  $IO_3^-$  formed
- (D)  $KMnO_4$  on heating does not give  $O_2$

## SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +3 If ONLY the correct option is chosen.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered)

*Negative Marks* : -1 In all other cases

7.

| List-I<br>(Colour of ppt.) |             | List-II<br>(From reactions in aq. solution) |                                                          |
|----------------------------|-------------|---------------------------------------------|----------------------------------------------------------|
| (P)                        | Red         | (1)                                         | $\text{AgNO}_3 + \text{NaBr} \rightarrow$                |
| (Q)                        | Green       | (2)                                         | $\text{Hg}_2^{2+} + \text{CrO}_4^{2-} \rightarrow$       |
| (R)                        | Pale yellow | (3)                                         | $\text{Ni}^{2+} + \text{NaOH} \rightarrow$               |
| (S)                        | White       | (4)                                         | $\text{Ca}^{2+} + \text{C}_2\text{O}_4^{2-} \rightarrow$ |

The correct option is :-

(A)  $\text{P} \rightarrow 1; \text{Q} \rightarrow 4; \text{R} \rightarrow 2; \text{S} \rightarrow 3$

(B)  $\text{P} \rightarrow 2; \text{Q} \rightarrow 3; \text{R} \rightarrow 4; \text{S} \rightarrow 1$

(C)  $\text{P} \rightarrow 2; \text{Q} \rightarrow 3; \text{R} \rightarrow 1; \text{S} \rightarrow 4$

(D)  $\text{P} \rightarrow 4; \text{Q} \rightarrow 2; \text{R} \rightarrow 1; \text{S} \rightarrow 3$

8.

| List-I<br>(Compounds) |                 | List-II<br>(Reagent) |                                                               |
|-----------------------|-----------------|----------------------|---------------------------------------------------------------|
| (P)                   | $\text{FeCl}_3$ | (1)                  | Insoluble in water as well as excess $\text{NH}_3$ solution   |
| (Q)                   | $\text{AgI}$    | (2)                  | Give orange ppt with $\text{K}_3[\text{Fe}(\text{CN})_6]$     |
| (R)                   | $\text{AgNO}_3$ | (3)                  | Give brown solution with $\text{K}_3[\text{Fe}(\text{CN})_6]$ |
| (S)                   | $\text{CaCl}_2$ | (4)                  | Give white ppt. with $\text{K}_4[\text{Fe}(\text{CN})_6]$     |

The correct option is :-

(A)  $\text{P} \rightarrow 4; \text{Q} \rightarrow 3; \text{R} \rightarrow 1; \text{S} \rightarrow 2$

(B)  $\text{P} \rightarrow 3; \text{Q} \rightarrow 1; \text{R} \rightarrow 2; \text{S} \rightarrow 4$

(C)  $\text{P} \rightarrow 3; \text{Q} \rightarrow 1; \text{R} \rightarrow 4; \text{S} \rightarrow 2$

(D)  $\text{P} \rightarrow 1; \text{Q} \rightarrow 3; \text{R} \rightarrow 4; \text{S} \rightarrow 2$

9.

| List-I<br>(Cation in aq. solution) |                  | List-II |                                                                 |
|------------------------------------|------------------|---------|-----------------------------------------------------------------|
| (P)                                | $\text{Cr}^{3+}$ | (1)     | Give white ppt with dil. HCl which soluble in hot water         |
| (Q)                                | $\text{Co}^{2+}$ | (2)     | Give black ppt with $\text{H}_2\text{S}$ and blue ppt with NaOH |
| (R)                                | $\text{Pb}^{2+}$ | (3)     | Give blue colour on adding conc. HCl                            |
| (S)                                | $\text{Cu}^{2+}$ | (4)     | Give yellow solution with NaOH + $\text{H}_2\text{O}_2$         |

The correct option is :-

(A)  $\text{P} \rightarrow 4; \text{Q} \rightarrow 3; \text{R} \rightarrow 1; \text{S} \rightarrow 2$

(B)  $\text{P} \rightarrow 4; \text{Q} \rightarrow 1; \text{R} \rightarrow 2; \text{S} \rightarrow 3$

(C)  $\text{P} \rightarrow 3; \text{Q} \rightarrow 1; \text{R} \rightarrow 2; \text{S} \rightarrow 4$

(D)  $\text{P} \rightarrow 1; \text{Q} \rightarrow 4; \text{R} \rightarrow 3; \text{S} \rightarrow 2$

10.

| List-I<br>(Cations) |                  | List-II<br>(Bead colour in oxidizing flame in cold condition) |        |
|---------------------|------------------|---------------------------------------------------------------|--------|
| (P)                 | $\text{Cu}^{2+}$ | (1)                                                           | Brown  |
| (Q)                 | $\text{Cr}^{3+}$ | (2)                                                           | Blue   |
| (R)                 | $\text{Mn}^{2+}$ | (3)                                                           | Green  |
| (S)                 | $\text{Ni}^{2+}$ | (4)                                                           | Violet |

The correct option is :-

(A)  $\text{P} \rightarrow 3; \text{Q} \rightarrow 2; \text{R} \rightarrow 1; \text{S} \rightarrow 4$

(B)  $\text{P} \rightarrow 2; \text{Q} \rightarrow 3; \text{R} \rightarrow 1; \text{S} \rightarrow 4$

(C)  $\text{P} \rightarrow 4; \text{Q} \rightarrow 3; \text{R} \rightarrow 2; \text{S} \rightarrow 1$

(D)  $\text{P} \rightarrow 2; \text{Q} \rightarrow 3; \text{R} \rightarrow 4; \text{S} \rightarrow 1$

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +3 If ONLY the correct numerical value is entered.

*Zero Marks* : 0 In all other cases.

1. Consider following cations in aqueous solution :-  
 $\text{Pb}^{2+}$ ,  $\text{Hg}_2^{2+}$ ,  $\text{Ag}^+$ ,  $\text{Cr}^{3+}$ ,  $\text{Al}^{3+}$ ,  $\text{Zn}^{2+}$ ,  $\text{Mn}^{2+}$ ,  $\text{Co}^{2+}$

Calculate value of  $\left( \frac{x+y}{z} \right)$ .

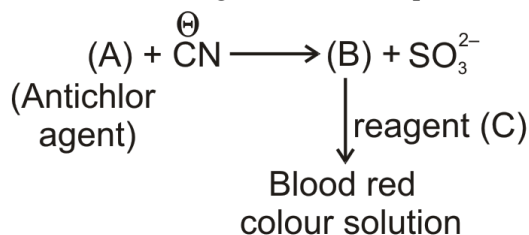
where

x = Total number of cation/s which give white precipitate of sulphide with  $\text{Na}_2\text{S}_{(\text{aq})}$

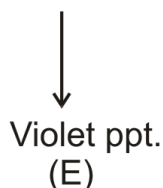
y = Total number of cation/s which give white precipitate with dil. HCl

z = Total number of cation/s which gives clear solution in excess NaOH solution

2. Consider the following reaction/s in aqueous solution.



complex compound (D) + aq. solution of A



if total number of unpaired electron present in complex compound (D) is x

if total number of geometrical isomer show by compound (E) than calculate x + y.

3. Total number of nitrate salt which give paramagnetic gas on thermal decomposition  
 $\text{LiNO}_3$ ,  $\text{NaNO}_3$ ,  $\text{KNO}_3$ ,  $\text{RbNO}_3$ ,  $\text{Pb}(\text{NO}_3)_2$ ,  $\text{Mg}(\text{NO}_3)_2$ ,  $\text{NH}_4\text{NO}_3$

4. Total number of reaction/s in which green ppt. obtain in aqueous solution.
- $\text{Ni}^{2+} + \text{DMG}^{-1} \rightarrow$
  - $\text{CuSO}_4 + [\text{Fe}(\text{CN})_6]^{3-} \rightarrow$
  - $\text{Hg}_2^{2+} + \text{I}^- \rightarrow$
  - $\text{Ni}^{2+} + \text{I}^- \rightarrow$
  - $\text{Ag}^+ + \text{PO}_4^{3-}$
  - $\text{Ag}^+ + \text{CO}_3^{2-} \rightarrow$
  - $\text{Fe}^{3+} + \text{CH}_3\text{COO}^\ominus \rightarrow$
  - $\text{Fe}^{3+} + \text{S}^{2\ominus} \rightarrow$
5. Which of the following reagent/s give  $\text{N}_2$  gas with  $\text{NO}_2^\ominus(\text{aq.})$   
 Sulphamic acid,  $\text{C}_2\text{H}_5\text{NH}_2$ , heating with  $\text{NH}_4\text{Cl}$ , Urea, Thiourea,  $\text{Zn} + \text{NaOH}$
6. Total number of elements/ions which give  $\text{H}_2$  with dilute  $\text{HCl}$ .  
 $\text{Na}$ ,  $\text{Zn}$ ,  $\text{Cu}$ ,  $\text{Ag}$ ,  $\text{Cr}^{2+}$ ,  $\text{Mn}^{2+}$ ,  $\text{Ti}^{2+}$ ,  $\text{Fe}^{2+}$ ,  $\text{Co}^{2+}$
7. 1.  $\text{XeF}_4 + \text{SbF}_5 \rightarrow \text{product (P)}$   
 2.  $\text{XeF}_4 + \text{O}_2\text{F}_2 \rightarrow \text{product (Q)} + \text{paramagnetic gas (G)}$   
 Consider above reaction and select total number of correct statement/s.
- In product (Q) formation  $\text{O}_2\text{F}_2$  act as Fluorinating agent
  - Product (Q) in solid state contain distorted octahedral structure
  - When product (Q) undergoes complete hydrolysis than it give explosive compound
  - Product (P) is ionic compound and Xe atom is anionic part
  - Product (P) is ionic compound and Xe atom in cationic part
  - In product (P) no ionic product is formed
  - In product (P) Xe atom has  $\text{sp}^3\text{d}^2$  hybridization
8. Consider following compound/s :-  
 $\text{XeF}_2$ ,  $\text{XeF}_4$ ,  $\text{XeF}_6$ ,  $\text{XeOF}_4$ ,  $\text{XeO}_3$ ,  $\text{XeO}_2\text{F}_2$   
 if (A) is/are total number of compound/s which give oxygen when react with water  
 (B) is/are total number of compound/s which show non-redox reaction when added in water  
 (C) is/are total number of planar compound/s  
 then calculate  $\frac{(A+B)}{C}$

**PART-3 : MATHEMATICS**  
**SECTION-I (i) : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

- 
1. If the real values of 'a' for which the equation  $\left(x - \log_2(a^6 + 3a^2 + 3a^4 + 1)\right)(x - 1) + 1 = 0$  has integral roots are  $a_1$  and  $a_2$ , then
    - (A)  $a_1 + a_2 = 0$
    - (B)  $a_1^2 + a_2^2 = 2$
    - (C)  $|a_1 - a_2| = 2$
    - (D)  $|a_1 - a_2| = 0$
  2. Possible set of values of 'a' for which the equation  $(x^2 + x)^2 + a(x^2 + x) + 4 = 0$  have all four imaginary roots is/are
 

|                                           |                                         |
|-------------------------------------------|-----------------------------------------|
| (A) $\left(\frac{65}{4}, \infty\right)$   | (B) $\left(-4, \frac{65}{4}\right)$     |
| (C) $a \in \left(-4, \frac{65}{8}\right)$ | (D) $\left(\frac{65}{8}, \infty\right)$ |

3. If  $a, b \in \mathbb{R}$  satisfy the equation  $a^3 + 24ab + 64 = 8b^3$ , then roots  $\alpha, \beta$  ( $\alpha < \beta$ ) of equation  $x^2 + bx + a = 0$  satisfies (provided  $0 < b < 3$ )
- (A)  $\alpha \in (-4, -2)$  (B)  $\beta \in (-1, 2)$   
 (C)  $\alpha \in (-4, -2]$  (D)  $\beta \in (-4, -2)$
4. If equation  $\left(\log_{3a}^{(a^2-4)}\right) x^2 + bx + 1 = 0$  doesn't have any real roots (provided all coefficient are real), then point possible set of values of  $a$  contains
- (A)  $(-2, -1)$  (B)  $(1, 2)$   
 (C)  $(3, 2\sqrt{5})$  (D)  $(\sqrt{5}, \infty)$
5. If the equation  $e^{4x} - ae^{3x} + 11e^{2x} - ae^x + 1 = 0$  has two distinct positive roots then  $a$  may take the value/s
- (A) 6 (B)  $\frac{13}{2}$   
 (C)  $\frac{25}{4}$  (D)  $\frac{19}{3}$
6. If  $S_n$  denotes the sum to  $n$  terms of series ( $1 \leq n \leq 9$ )  
 $1 + 22 + 333 + \dots + \underbrace{99\dots9}_{9 \text{ times}}$   
 then for  $n \geq 2$
- (A)  $S_n - S_{n-1} = \frac{1}{9}(10^n - n^2 + n)$   
 (B)  $S_n = \frac{1}{9}(10^n - n^2 + 2n - 2)$   
 (C)  $9(S_n - S_{n-1}) = n(10^n - 1)$   
 (D)  $S_3 = 356$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +3 If ONLY the correct option is chosen.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered)

*Negative Marks* : -1 In all other cases

7. If  $\log_3 2 = a$ ,  $\log_4 5 = b$  and  $\log_5 6 = c$  then match the following columns

|     | List-I                                             |     | List-II                     |
|-----|----------------------------------------------------|-----|-----------------------------|
| (P) | $\log_{10} 135$                                    | (1) | $\frac{3 + 2ab}{a(1 + 2b)}$ |
| (Q) | $\log_4 36$                                        | (2) | $\frac{2bc + 1}{2bc + 2}$   |
| (R) | $\log_{24} 12$                                     | (3) | $\frac{a + 1}{a}$           |
| (S) | $2\log_{10} 3 + \frac{1 + \log_5 3}{1 + \log_5 2}$ | (4) | $2bc$                       |

- (A)  $P \rightarrow 1; Q \rightarrow 3, 4; R \rightarrow 2; S \rightarrow 1$  (B)  $P \rightarrow 3, 4; Q \rightarrow 1; R \rightarrow 1; S \rightarrow 2$   
 (C)  $P \rightarrow 2; Q \rightarrow 3, 4; R \rightarrow 4; S \rightarrow 1$  (D)  $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 3, 4$

8. If  $x, y, z, u > 0$  are distinct real numbers in G.P. and  $a, b, c, d$  are real & distinct, non-zero numbers in A.P., then match the correct answers.

|     | List-I                                                                           |     | List-II        |
|-----|----------------------------------------------------------------------------------|-----|----------------|
| (P) | $a - \log \frac{x}{y^3}, b - \log \frac{y}{z^3}, c - \log \frac{z}{u^3}$ will be | (1) | only in A.P.   |
| (Q) | $(b - a)xy, (c - b)yz, (d - c)zu$ will be                                        | (2) | only in G.P.   |
| (R) | $\frac{a(b + c)}{bc}, \frac{b(c + a)}{ca}, \frac{c(a + b)}{ab}$ will be          | (3) | only in H.P.   |
| (S) | $(a^2 - b^2)(x^2 - y^2), (b^2 - c^2)(y^2 - z^2), (c^2 - d^2)(z^2 - u^2)$ will be | (4) | only in A.G.P. |
|     |                                                                                  | (5) | none of above  |

- (A)  $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$  (B)  $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$   
 (C)  $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$  (D)  $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 5; S \rightarrow 1$

9.

|     | List-I                                                                                                                                                                       |     | List-II                                          |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------------------------------------------------|
| (P) | If $p^{\text{th}}, q^{\text{th}}$ and $r^{\text{th}}$ terms of a G.P. (common ratio $\neq 1$ ) are in G.P. then $p, q, r$ will be in                                         | (1) | Arithmetic progression                           |
| (Q) | If $a, b, c, d$ are non-zero real numbers such that $(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) < (ab + bc + cd)^2$ and $c(a - b) = a(b - c)$ , then $a, b, c, d$ will be in         | (2) | Geometric progression                            |
| (R) | If $a, b, c, d$ are four different positive numbers in G.P. then $\frac{a(2 \log a + \log b)}{b}, \frac{b(2 \log b + \log c)}{c}, \frac{c(2 \log c + \log d)}{d}$ will be in | (3) | Harmonic progression                             |
| (S) | If $a_1, a_2, a_3$ are in A.P., $a_2, a_3, a_4$ are in G.P. and $a_3, a_4, a_5$ are in H.P. then $a_1, a_3, a_5$ are always in                                               | (4) | Arithmetic geometric progression                 |
|     |                                                                                                                                                                              | (5) | A progression, whose all the terms are identical |

(A)  $P \rightarrow 2, 5; Q \rightarrow 3, 4; R \rightarrow 3, 5; S \rightarrow 1, 5$

(B)  $P \rightarrow 1, 5; Q \rightarrow 3, 5; R \rightarrow 3, 4; S \rightarrow 2, 5$

(C)  $P \rightarrow 1, 4; Q \rightarrow 1, 2, 3, 4, 5; R \rightarrow 1, 4; S \rightarrow 2, 4$

(D)  $P \rightarrow 2, 4; Q \rightarrow 1, 4; R \rightarrow 2, 3; S \rightarrow 1, 5$

10.

|     | List-I                                                                                                                                          |     | List-II                                                 |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------|
| (P) | If $A = \left\{ x \mid \log_{\frac{2}{3}}(x-2)^2 \geq 0; x \in \mathbb{R} \right\}$ , then possible subset of A is/are                          | (1) | $\left( 0, \frac{1}{3} \right)$                         |
| (Q) | Subset of interval in which 'a' lies if $x^2 - (\alpha - \beta)x + (1 - \alpha - \beta) = 0$ has distinct real roots for all real $\beta$ , is  | (2) | $(1, 2)$                                                |
| (R) | The interval in which 'b' lies, if the roots of $x^3 - 6x^2 + 2ax - 4b = 0$ , where $a \in (5, 6)$ , are in A.P., is                            | (3) | $\left( 2, \frac{7}{2} \right)$                         |
| (S) | Subset of set of values of $\lambda$ for which both roots of equation $x^2 - 6\lambda x + 2 - 2\lambda + 9\lambda^2 = 0$ are greater than 3, is | (4) | $(-\infty, 1) \cup \left( \frac{11}{9}, \infty \right)$ |

(A)  $P \rightarrow 1; Q \rightarrow 2, 3; R \rightarrow 2; S \rightarrow 4$

(B)  $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

(C)  $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$

(D)  $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

## SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +3 If ONLY the correct numerical value is entered.

*Zero Marks* : 0 In all other cases.

- Find the sum of infinite terms of a decreasing G.P. whose first term is equal to  $p + 2$  and the common ratio is  $\frac{2}{q}$ , where  $p$  is the least value of the product of the roots of the equation  $(m^2 + 1)x^2 - 3x + (m^2 + 1)^2 = 0$  and  $q$  is the greatest value of the sum of its roots.
- If the sum of the series  $\frac{x}{1-x^2} + \frac{x^2}{1-x^4} + \frac{x^4}{1-x^8} + \dots$  upto 10 terms is  $\frac{1}{1-x} - \frac{1}{1-x^k}$ , then the value of  $k$  must be
- Evaluate the sum  $\sum_{k=1}^{\infty} \frac{6^k}{(3^k - 2^k)(3^{k+1} - 2^{k+1})}$ .
- If the range of values of  $k$  for which there is at least one common solution of the inequalities  $x^2 + 4kx + 3k^2 > 1 + 2k$  and  $x^2 + 2kx \leq 3k^2 - 8k + 4$ , is  $(-\infty, a) \cup (b, \infty)$ , then find the value of  $(a + b)$ .
- When  $x^{100}$  is divided by  $x^2 - 3x + 2$ , then remainder is  $(2^{k+1} - 1)x - 2(2^k - 1)$  where  $k$  is a numerical quantity. Then  $k$  must be
- Given that  $\alpha, \gamma$  are roots of the equation  $Ax^2 - 4x + 1 = 0$  and  $\beta, \delta$  the roots of the equation  $Bx^2 - 6x + 1 = 0$ . If the value of  $A$  and  $B$  such that  $\alpha, \beta, \gamma$  and  $\delta$  are in H.P. are  $\lambda$  and  $\lambda + 5$  respectively, then the numerical quantity  $\lambda$  must be
- If  $a_1, a_2, a_3 \dots a_{100}$  are in H.P. then  $\sum_{i=1}^{99} \frac{a_i a_{i+1}}{a_1 a_{100}}$  equals
- If  $a, b, c$  are positive and  $a + b + c = 1$ , then find minimum value of  $\left(\frac{1}{a} - 1\right) \left(\frac{1}{b} - 1\right) \left(\frac{1}{c} - 1\right)$  equals

---

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. ....

I have read all the instructions  
and shall abide by them.

\_\_\_\_\_  
Signature of the Candidate

I have verified the identity, name and Form  
number of the candidate, and that question  
paper and ORS codes are the same.

\_\_\_\_\_  
Signature of the Invigilator

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E-24/24

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